BANK MARKETING– Data Analysis

Business Objective

The bank possesses a large volume of customer demographic, financial, and campaign interaction data but lacks actionable insights to improve marketing effectiveness. There is limited visibility into which customer segments are most likely to subscribe to term deposits, how marketing campaigns influence customer decisions, and which channels and contact strategies deliver the best results.

This project aims to conduct an end-to-end analysis of the Bank Marketing dataset to:

1. Identify key drivers influencing term deposit subscriptions.
2. Understand customer demographic and financial behaviour.
3. Evaluate the effectiveness of marketing campaigns and contact strategies.
4. Assess the impact of call duration, contact frequency, and timing on conversion.
5. Provide data-driven recommendations to improve campaign success rates and customer targeting.

Problem Statement

The organization seeks to understand the key factors affecting customer response to marketing campaigns. Despite having detailed customer and campaign data, decision-makers lack clear insights into which customer profiles are most responsive, which campaign strategies work best, and where inefficiencies exist.

The goal is to analyze historical marketing campaign data to identify trends, discover improvement areas, and provide actionable recommendations to increase term deposit subscription rates.

Phase 1: Business Understanding

Business Questions

• Which customer segments are most likely to subscribe to a term deposit?
• How do demographic factors (age, job, marital status, education) influence subscription?
• How does call duration impact campaign success?
• Which contact methods and months yield higher conversion rates?
• How does previous campaign outcome affect future subscription?

Phase 2: Data Understanding

Data Exploration

Q1. Steps Taken During Data Exploration

The following steps were performed during data exploration:

1. Reviewed dataset structure and attributes.
2. Identified categorical and numerical variables.
3. Checked for missing and unknown values.
4. Verified data types and value ranges.
5. Analyzed target variable distribution (term deposit subscription).
6. Examined customer demographics and financial indicators.
7. Reviewed campaign-related variables (calls, duration, contact type).
8. Identified outliers in call duration and number of contacts.
9. Performed initial aggregations and pivot analysis.
10. Checked class imbalance in the target variable.

Q2. Insights Gained During Data Exploration

The following insights were identified:

• Majority of customers did not subscribe to term deposits.
• Longer call durations are strongly associated with higher subscription rates.
• Customers with higher education levels are more responsive.
• Retired and student segments show higher conversion rates.
• Cellular contact method is more effective than telephone.
• Fewer contact attempts generally lead to better outcomes.
• Previous campaign success significantly increases conversion probability.
• Certain months show higher campaign effectiveness.

Q3. Calculated Columns Created

The following calculated columns were created:

1. Age Group
   (e.g., 18–30, 31–45, 46–60, 60+)
2. Call Duration (Minutes)
   Call Duration (seconds) ÷ 60
3. Contact Frequency Category
   Low, Medium, High based on number of contacts
4. Campaign Response Rate
   Successful Subscriptions ÷ Total Contacts
5. Customer Risk Indicator
   Based on loan and default status
6. Previous Campaign Outcome Flag
   Success / Failure / No Contact

Q4. KPIs Created and Analysis

Key Performance Indicators (KPIs)

• Total Customers Contacted
• Total Subscriptions
• Subscription Rate (%)
• Average Call Duration
• Average Contacts per Customer
• Campaign Success Rate
• Conversion Rate by Segment

Insights from KPIs

• Higher call duration significantly improves subscription likelihood.
• Excessive contact attempts reduce customer response.
• Cellular contact leads to higher conversion rates.
• Educated and financially stable customers respond better.
• Targeted campaigns outperform mass marketing.

Phase 3: Data Preparation

Data Cleaning

1. Handling Dimension and Measure Names

• Standardized column naming conventions.
• Removed special characters and inconsistencies.
• Renamed ambiguous fields for clarity.

2. Handling Missing Values

• Treated "unknown" values as missing.
• Replaced or categorized missing demographic data.
• Removed records with insufficient campaign information.

3. Data Type Formatting

• Converted categorical variables to text.
• Converted numeric variables to appropriate formats.
• Ensured target variable is binary (Yes/No).

Feature Engineering

• Created Age Groups and Customer Segments.
• Converted call duration to minutes.
• Created contact frequency buckets.
• Derived response indicators from campaign outcomes.
• Created month and quarter features from contact date.

Data Integration and Formatting

• Ensured consistency between customer and campaign data.
• Removed duplicate records.
• Standardized categorical values.
• Prepared data for analysis and visualization.

Phase 3.1: Data Exploration and Analysis

Analysis Performed

- Subscription rate by customer segment.
- Impact of call duration on subscription.
- Contact method and timing analysis.
- Effect of previous campaign outcomes.
- Age, job, and education-based conversion analysis.
- Campaign efficiency and contact optimization.
- Class imbalance and response trend analysis.

Key Findings and Recommendations

Key Findings

- Longer, meaningful calls increase subscription probability.
- Retired and student segments have higher response rates.
- Cellular contact method performs best.
- Repeated calls reduce customer interest.
- Previous campaign success is a strong predictor of future response.

Recommendations

1. Focus marketing efforts on high-response customer segments.
2. Optimize call duration rather than increasing call frequency.
3. Prioritize cellular contact over telephone.
4. Avoid excessive follow-up calls.
5. Use previous campaign outcomes for targeted marketing.